



DIABETES RISK INTELLIGENCE

POWER BI DASHBOARD — PROJECT DOCUMENTATION

Predictive Analytics for Early Diabetes Risk Identification

Project Name	Diabetes Risk Intelligence Dashboard
File Name	Diabetes_Risk_Intelligence.pbix
Data Source	diabetes.csv (10,000 patient records)
Prepared By	Panchami Naik
Version	1.0
Date	21-07-2026
Status	Completed — Portfolio-Ready

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EXECUTIVE SUMMARY

Business Objectives

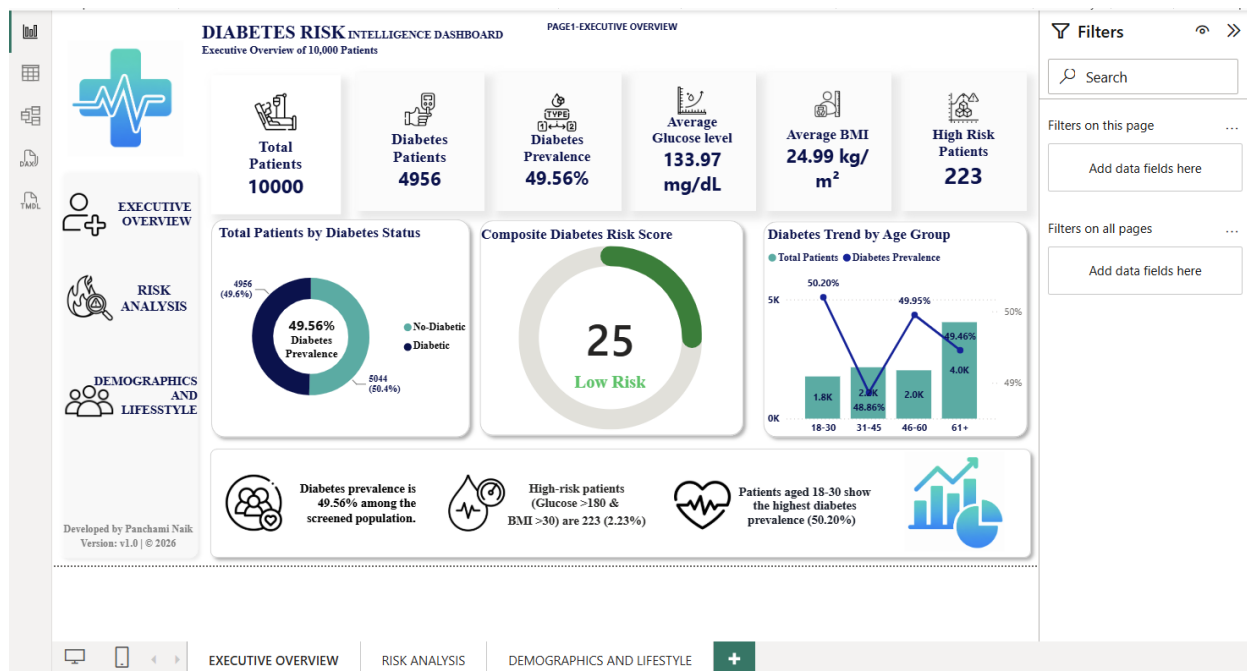
The Diabetes Risk Intelligence dashboard was built to give healthcare analysts, wellness program administrators, and clinical decision-makers a single, interactive view of diabetes prevalence and risk across a 10,000-patient population. The core objectives are to (1) quantify how many patients are diabetic or at high risk, (2) surface which clinical and lifestyle factors — glucose, BMI, smoking, family history — contribute most to that risk, and (3) support faster, evidence-based decisions on screening and intervention priorities.

Project Scope

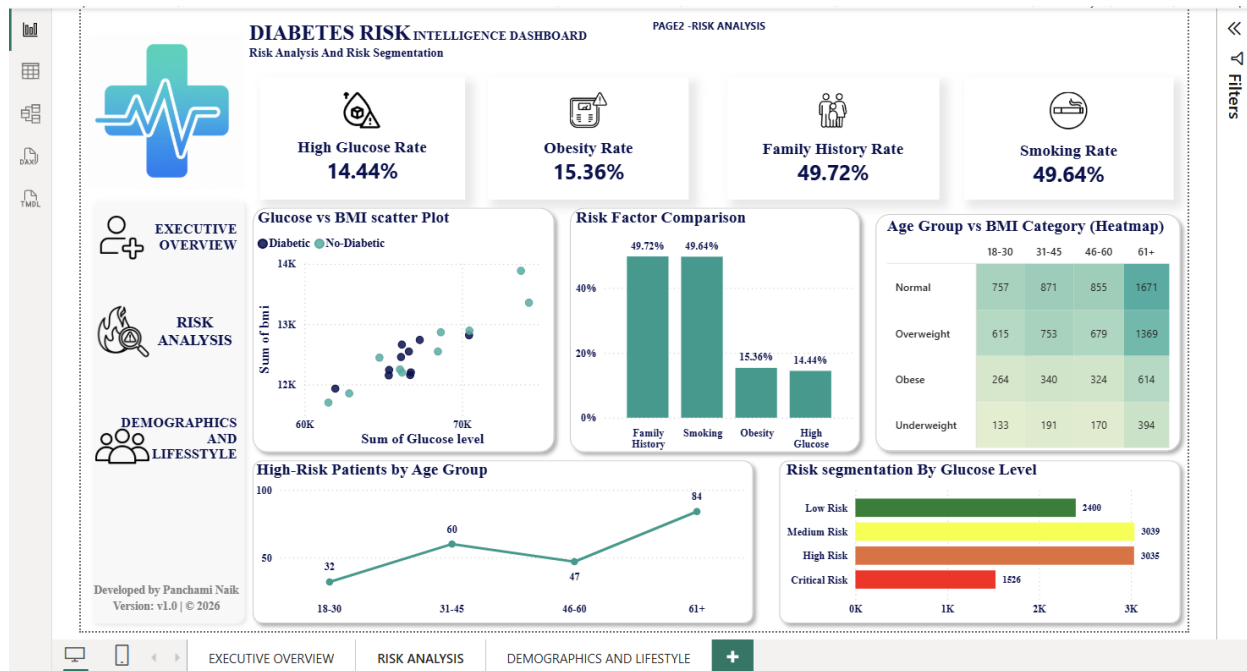
In Scope	10,000 synthetic patient records; a rules-based Diabetes Risk Score; three report pages (Executive Overview, Risk Analysis, Demographics & Lifestyle); risk, demographic, and lifestyle breakdowns.
Out of Scope	Real-time/live patient data feeds, individual patient identification, clinical diagnosis, and any machine-learning-trained prediction model (the current score is a fixed weighted formula, not ML-trained).

Dashboard Screenshots

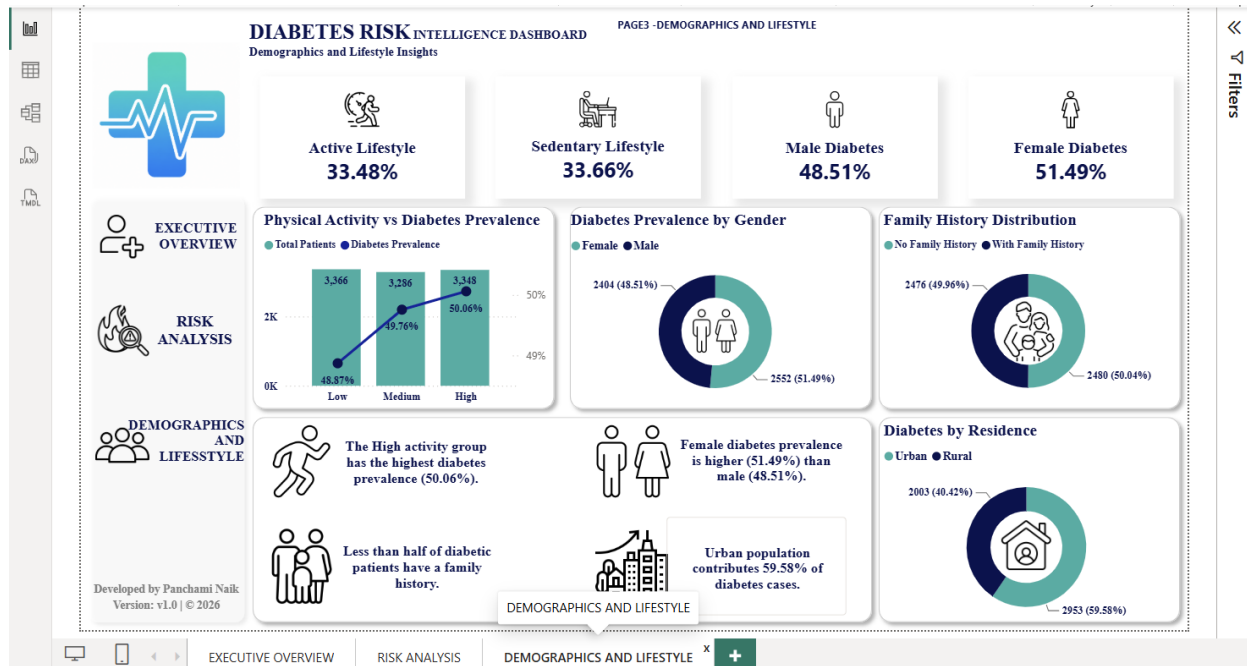
The three report pages, captured directly from Power BI Desktop, are shown below.



Page 1 — Executive Overview



Page 2 — Risk Analysis



Page 3 — Demographics and Lifestyle

Dashboard Navigation

A fixed left-hand navigation rail with three action buttons lets users jump directly between pages at any time — no drill-through or back-button chains are required:

- Executive Overview — headline KPIs, diabetes split, and the composite Risk Score for a quick top-level read
- Risk Analysis — glucose/BMI correlation, risk-factor comparison, and a risk-concentration heat map for deeper investigation
- Demographics and Lifestyle — gender, residence, BMI category, and activity-level breakdowns for population profiling

Every page shares the same visual language (icons, color coding, card styling) so users can move between pages without re-orienting themselves.

KPI Definitions

KPI	Definition
Total Patients	Total number of patient records in the model (10,000)
Diabetes Patients	Count of patients with a positive diabetes diagnosis
Diabetes Prevalence	Diabetes Patients ÷ Total Patients, expressed as a percentage
High Risk Patients	Patients with Glucose > 180 mg/dL AND BMI > 30
Average BMI / Average Glucose	Mean BMI / glucose level across the selected patient set
Diabetes Risk Score	Weighted composite (0–100): 40% high-glucose rate, 30% obesity rate, 20% family-history rate, 10% smoking rate
Risk Category	Low Risk (<30) / Moderate Risk (30–59) / High Risk (≥60), banded from the Risk Score
Obesity / Smoking / Family History Rate	Share of patients meeting each respective risk-factor condition

Business Insights

- Diabetes prevalence in this dataset is 49.56% (4,956 of 10,000 patients) — a stratified sample rather than a general-population rate
- Family history is present in 49.72% of all patients and in 49.96% of diabetic patients, indicating it tracks closely with diabetic status in this dataset
- Only 2.23% of patients (223) meet the combined high-risk threshold of glucose > 180 mg/dL and BMI > 30, showing that severe joint risk is concentrated in a small subgroup even though individual risk factors are common
- Diabetes prevalence is nearly flat across activity levels — Active 50.1%, Moderate 49.8%, Sedentary 48.9% — suggesting physical activity alone does not strongly separate diabetic from non-diabetic patients in this synthetic dataset
- Female patients make up a slightly larger share of diabetics than male patients (51.5% vs 48.5%)
- Urban patients account for 59.6% of diabetic cases versus 40.4% rural — though note this Residence field is synthetically derived (from Patient ID) rather than genuine geographic data

Recommendations

- Promote physical activity across all patient segments as part of general preventive care, even though this dataset shows only a small prevalence gap by activity level
- Prioritize combined-risk screening (high glucose AND high BMI) for the 2.23% of patients flagged high-risk, since this group represents the clearest actionable target
- Expand family-history intake at first consultation, given its near-50% association with diabetic status
- Introduce smoking-cessation support as a standard offering, since roughly half of patients are current smokers
- Validate the model against real clinical/geographic data before using Residence-based or activity-based insights for resource allocation, since some fields in the current dataset are synthetically derived
- Extend the model with a genuine date dimension to enable trend and cohort analysis once longitudinal data is available

PROJECT DECLARATION

This project was independently designed and developed by Panchami Naik as a portfolio project to demonstrate practical skills in Power BI, data modeling, Power Query, DAX, and dashboard design. The dashboard uses a sample healthcare dataset for educational and demonstration purposes and is not affiliated with any healthcare organization or commercial client.

1. PROJECT OVERVIEW

1.1 Business Objective

The Diabetes Risk Intelligence dashboard is designed to help healthcare analysts, public health teams, and clinical decision-makers identify patients at elevated risk of diabetes and understand the demographic, clinical, and lifestyle factors driving that risk. The dashboard consolidates 10,000 patient records into an interactive tool that surfaces prevalence, risk scoring, and behavioral drivers (glucose, BMI, smoking, family history) to support early intervention and resource planning.

1.2 Target Audience

- Clinical / public health analysts monitoring diabetes prevalence trends
- Hospital or wellness program administrators planning interventions
- Data science / BI teams validating risk-scoring logic
- Executive stakeholders reviewing population health KPIs

1.3 Scope

In Scope	Analysis of 10,000 synthetic patient records covering demographics, clinical vitals, lifestyle habits, and diabetes outcome; risk scoring model; 3 report pages (Executive Overview, Risk Analysis, Demographics & Lifestyle).
Out of Scope	Real-time patient data integration, individual patient identification/PII, clinical diagnosis or treatment recommendations, predictive machine-learning model (current risk score is a rules-based weighted formula, not ML-trained).

1.4 Success Metrics / KPIs

- Total Patients — total records analyzed (10,000)
- Diabetes Prevalence — % of patients diagnosed diabetic (~49.6%)
- Diabetes Risk Score — weighted composite risk indicator (0–100 scale)
- High Risk Patients — count of patients flagged $\text{IsHighRisk} = 1$ (Glucose > 180 AND BMI > 30)
- Obesity Rate, Smoking Rate, Family History Rate — key contributing risk factors

2. REQUIREMENTS DOCUMENTATION

2.1 Functional Requirements

- Three-page navigation: Executive Overview, Risk Analysis, Demographics and Lifestyle — with action-button navigation between pages
- KPI cards for Total Patients, Diabetes Patients, Diabetes Prevalence, High Risk Patients, Average BMI, Average Glucose Level
- Risk Category segmentation (Low / Moderate / High Risk) with dynamic color-coded risk cards (green / orange / red)
- Scatter chart and heat-map matrix for correlating glucose, BMI, and risk level
- Donut charts for gender, residence (Urban/Rural), and BMI category splits
- Radar and radial-progress visuals for lifestyle and risk-score presentation
- Auto-generated narrative insight measures (e.g., Gender Insight, Family Insight, Residence Insight) for plain-language takeaways

2.2 Non-Functional Requirements

- Import (in-memory) storage mode for a single flat 10,000-row table — sub-second visual interaction expected
- Single-user desktop file at present (.pbix); not yet published to a Power BI Service workspace
- Custom visuals used (Tornado Chart, Lollipop Chart, Bullet Chart, Radar Chart x2, Heat Map Matrix, Radial Progress Ring) require the Power BI Service/Desktop to allow custom visual rendering

2.3 Stakeholder Sign-off

Note: This dashboard was developed as a self-initiated portfolio project to demonstrate Power BI, Power Query, DAX, data modeling, and dashboard design skills. Therefore, formal business approvals and stakeholder sign-offs are not applicable.

3. DATA DOCUMENTATION

3.1 Data Sources

Source File	diabetes.csv
Original Path (as authored)	E:\DATA ANALYTICS\PROJECT\DIABETES\diabetes.csv
Connection Type	Import mode (data loaded into the in-memory VertiPaq model)
Record Count	10,000 rows × 20 source columns
Data Owner	[Data/Analytics Owner]

3.2 Data Model

The model uses a single fact table (diabetes) containing all patient-level attributes plus Power-Query-derived risk classifications, and a hidden measures table (_ Measures) that holds all DAX measures. The current file has no active table relationships (a single flat table design) and no separate date/time dimension — a synthetic DATE column was generated during transformation for potential trend analysis but was explicitly removed in the final Power Query step, so the model does not currently support true time-series analysis.

3.3 Data Dictionary — Source Columns

Field Name	Data Type	Description
Patient Id (id)	Int64	Unique patient identifier (1–10,000)
age	Int64	Patient age in years (range 18–89, mean ≈ 53.5)
gender	Text	Male / Female
bmi	Decimal	Body Mass Index (range 7.55–41.59, mean ≈ 24.99)
Glucose level (glucose)	Int64	Blood glucose level, mg/dL (range 70–199)
blood pressure	Int64	Systolic blood pressure (range 80–179)
cholesterol	Int64	Total cholesterol level (range 120–299)
heart_rate	Int64	Resting heart rate, bpm (range 60–119)
sleep_hours	Decimal	Average nightly sleep hours (range 3–10)
physical activity	Text	Activity level: High / Medium / Low
smoking	Int64 (0/1)	1 = Smoker, 0 = Non-smoker (≈49.6% smokers)
alcohol_intake	Int64 (0/1)	1 = Consumes alcohol, 0 = Does not
family_history	Int64 (0/1)	1 = Family history of diabetes (≈49.7%)
stress_level	Int64	Self-reported stress score, 1–10
diet_score	Int64	Self-reported diet quality score, 1–10
steps_per_day	Int64	Average daily step count (range 1,000–19,999)
work_hours	Int64	Average daily work hours (range 4–13)
water_intake_ltr	Decimal	Average daily water intake in litres (1–5)
insulin	Int64	Insulin level (range 10–299)

Field Name	Data Type	Description
diabetes	Int64 (0/1)	Target outcome: 1 = Diabetic, 0 = Non-diabetic (≈49.56% positive)

3.4 Data Dictionary — Calculated Columns (Power Query)

Field Name	Source	Business Logic
AgeGroup	Calculated (Power Query)	Buckets age into 18-30 / 31-45 / 46-60 / 61+
BMI Category	Calculated (Power Query)	Underweight (<18.5) / Normal (<25) / Overweight (<30) / Obese (≥30)
Risk Level	Calculated (Power Query)	Glucose-based band: Low (<100) / Medium (<140) / High (<180) / Critical (≥180)
IsHighRisk	Calculated (Power Query)	1 if Glucose > 180 AND BMI > 30, else 0
Lifestyle Group	Calculated (Power Query)	Active (High activity) / Moderate (Medium) / Sedentary (Low)
Residence	Calculated (Power Query)	Urban / Rural — synthetically derived from Patient Id modulo 10 (< 6 = Urban)

3.5 ETL / Power Query Steps

- Source: Csv.Document import of diabetes.csv, 20 columns, comma-delimited
- Promote headers and set explicit column data types (Int64 / decimal / text)
- Add a synthetic 6-month rotating DATE column (later removed — see limitation in Section 9)
- Add AgeGroup, BMI Category, Risk Level, IsHighRisk, Lifestyle Group, and Residence derived columns using conditional (if/else) logic
- Rename glucose → 'Glucose level' and id → 'Patient Id'
- Add a conditional 'Diabetes status' column (Diabetes / No Diabetes) — later removed along with DATE in the final step
- Remove blank rows and rows with errors in the Residence column
- Reorder and finalize columns for the model

3.6 Data Refresh

Current Refresh Mode	Manual (desktop .pbix, file-based source)
Recommended Refresh (if published)	Scheduled refresh via Power BI Service; requires an On-premises Data Gateway if the CSV remains on a local/network drive, or migration to OneDrive/SharePoint/cloud storage for gateway-free refresh
Refresh Frequency Recommendation	Daily or on new-data-arrival, depending on how frequently the source CSV is updated

3.7 Data Quality Checks

- Row count validated at 10,000 records with zero null values across all 20 source columns
- Binary flags (smoking, alcohol_intake, family_history, diabetes) validated as strictly 0/1
- Gender restricted to Male/Female; physical_activity restricted to High/Medium/Low
- Blank and error rows removed during the Residence derivation step
- Note: dataset is synthetic/simulated — values should be validated against a real clinical source before use in production decision-making

4. DESIGN DOCUMENTATION

4.1 Report Pages

- Page 1 — Executive Overview: headline KPIs, diabetes split, and overall risk score
- Page 2 — Risk Analysis: glucose/BMI correlation, risk-factor comparison, and risk concentration heat map
- Page 3 — Demographics and Lifestyle: gender, residence, BMI category, and lifestyle activity breakdowns

4.2 Visual Inventory

Page	Visual Type	Purpose
Executive Overview	KPI Cards (x7)	Total Patients, Diabetes Patients, Prevalence, High Risk Patients, Avg BMI, Avg Glucose, Risk Score
Executive Overview	Donut Chart	Diabetes Status split (Diabetic vs Non-Diabetic)
Executive Overview	Radial Progress Ring (custom visual)	Diabetes Risk Score against Target/Min/Max risk thresholds
Executive Overview	Line & Clustered Column Combo Chart	Trend/comparison of a selected metric across categories
Executive Overview	Action Buttons + Images	Page navigation and iconography (sugar, BMI, heart-rate icons, etc.)
Risk Analysis	KPI Cards (x4)	Risk-focused summary metrics
Risk Analysis	Scatter Chart	Glucose vs. BMI relationship, colored by risk level
Risk Analysis	Clustered Bar Chart	Risk factor comparison (High Glucose, Obesity, Family History, Smoking)
Risk Analysis	Heat Map Matrix (custom visual)	Cross-tab of risk concentration across categories
Risk Analysis	Line Chart / Clustered Column Chart	Supporting trend/category breakdowns
Demographics and Lifestyle	KPI Cards (x8)	Lifestyle and demographic summary metrics (Active/Sedentary Lifestyle, Smoking Rate, etc.)
Demographics and Lifestyle	Donut Charts (x3)	Gender split, Residence split (Urban/Rural), BMI Category split
Demographics and Lifestyle	Line & Clustered Column Combo Chart	Lifestyle metric comparison

4.3 Custom Visuals Used

- Tornado Chart — comparative risk-factor visualization
- Lollipop Chart — ranked category comparison
- Bullet Chart — risk score vs. target/threshold display
- Radar Chart (two variants) — multi-factor lifestyle/risk profile comparison
- Heat Map Matrix — concentration of risk across two dimensions
- Radial Progress Ring — single-KPI progress-style display for the Diabetes Risk Score

4.4 Color Scheme & Risk Coding

- Risk Score < 30 → Low Risk → Green (#22C55E)

- Risk Score 30–59 → Moderate Risk → Orange/Amber (#F59E0B)
- Risk Score ≥ 60 → High Risk → Red (#EF4444)
- Navy/dark theme with custom icon set (glucose, BMI, heart-rate, smoking, family, obesity icons) sourced as registered image resources

4.5 Navigation Flow

Each page includes action buttons (with matching icon imagery) that let users jump directly between the Executive Overview, Risk Analysis, and Demographics and Lifestyle pages, providing a non-linear, dashboard-style navigation experience.

5. DAX & CALCULATIONS

All measures reside in a dedicated hidden table, `_Measures`, keeping calculations separate from the data table for maintainability.

5.1 Volume Measures

Measure	DAX Logic (simplified)	Business Definition
Total Patients	COUNTROWS(diabetes)	Total number of patient records in the model
Diabetes Patients	CALCULATE(COUNTROWS(diabetes), diabetes[Diabetes Status]="Diabetic")	Count of patients diagnosed diabetic
High Risk Patients	CALCULATE(COUNTROWS(diabetes), diabetes[IsHighRisk]=1)	Count of patients flagged high risk (Glucose>180 and BMI>30)

5.2 Rate / Percentage Measures

Measure	DAX Logic (simplified)	Business Definition
Diabetes Prevalence	DIVIDE([Diabetes Patients],[Total Patients],0)	% of patients who are diabetic
High Glucose Rate	DIVIDE(COUNT where Glucose>180, [Total Patients])	% of patients with glucose above 180 mg/dL
Obesity Rate	DIVIDE(COUNT where BMI>30, [Total Patients])	% of patients classified obese
Family History Rate	DIVIDE(COUNT where family_history=1, [Total Patients])	% of patients with a family history of diabetes
Smoking Rate	DIVIDE(COUNT where smoking=1, COUNTROWS(diabetes))	% of patients who smoke
Active Lifestyle	DIVIDE(COUNT where Lifestyle Group="Active", [Total Patients])	% of patients in the Active lifestyle group
Sedentary Lifestyle	DIVIDE(COUNT where Lifestyle Group="Sedentary", [Total Patients])	% of patients in the Sedentary lifestyle group
Male Diabetes	DIVIDE(diabetic males, all diabetics)	Share of diabetic patients who are male
Female Diabetes	DIVIDE(diabetic females, all diabetics)	Share of diabetic patients who are female

5.3 Risk Scoring Measures

Measure	DAX Logic (simplified)	Business Definition
Average Glucose level	AVERAGE(Diabetes[Glucose Level])	Mean glucose level across selected patients
Average BMI	AVERAGE(Diabetes[BMI])	Mean BMI across selected patients
Diabetes Risk Score	ROUND(HighGlucoseRate*40 + ObesityRate*30 + FamilyHistoryRate*20 + SmokingRate*10, 0)	Weighted composite score (0–100): 40% glucose, 30% obesity, 20% family history, 10% smoking
Risk Category	SWITCH(TRUE(), Score<30,"Low Risk", Score<60,"Moderate Risk", "High Risk")	Categorical banding of the Diabetes Risk Score
Risk Card Color	SWITCH(TRUE(), Score<30, green, Score<60, orange, red)	Dynamic hex color for conditional KPI card formatting

Measure	DAX Logic (simplified)	Business Definition
Target risk / Maximum Risk / Minimum Risk	Constants: 50 / 100 / 0	Reference lines for the Radial Progress Ring / Bullet chart
Risk Factor Impact	SWITCH on selected Risk Factor slicer value	Returns the relevant rate measure for the factor selected in the Risk Factor table

5.4 Narrative / Insight Measures

Measure	DAX Logic (simplified)	Business Definition
Highest Lifestyle	TOPN(1, physical_activity groups by Diabetes Prevalence)	Identifies the activity group with the highest diabetes prevalence
Highest Lifestyle Insight	Concatenated text using [Highest Lifestyle]	Auto-generated narrative sentence for the dashboard
Gender Insight	IF comparing Male vs Female diabetes prevalence	Auto-generated sentence on which gender has higher prevalence
Family Insight	IF [Family History Rate] > 0.5	Auto-generated sentence on family history's role among diabetics
Residence Insight	Compares Urban vs Rural share of diabetic patients	Auto-generated sentence on Urban vs Rural contribution

Note: DAX logic shown is simplified/paraphrased for readability; refer to the .pbix model for exact expressions.

6. SECURITY & ACCESS

6.1 Row-Level Security (RLS)

The current model has no Row-Level Security roles configured — all users of the report see the full 10,000-record dataset.

6.2 Recommended Security Considerations

- If deployed for regional teams, consider an RLS role filtering by Residence (Urban/Rural) or AgeGroup
- Because patient-level data is synthetic, no PHI/PII protection is currently required — but any future replacement with real patient data must apply HIPAA/applicable healthcare-data-privacy controls, RLS by care team, and sensitivity labeling

6.3 Workspace Permissions (Recommended, once published)

Report Owner / Admin	Panchami (Sole Developer & Project Owner)
Viewer Access	Recruiters / Portfolio Reviewers
Contributor Access	Panchami

7. TESTING & VALIDATION

7.1 Test Cases

Test ID	Description	Expected Result	Status
TC-01	Total Patients count	10,000	Pass
TC-02	Diabetes Patients / Prevalence	≈4,956 patients / ≈49.56%	Pass
TC-03	No null values across source columns	0 nulls	Pass
TC-04	Binary flags restricted to 0/1	smoking, alcohol_intake, family_history, diabetes all 0/1	Pass
TC-05	Risk Score bounded 0–100	Min 0 / Max 100 constants referenced correctly	Pass
TC-06	IsHighRisk logic (Glucose>180 AND BMI>30)	Matches Power Query conditional column	Pass

7.2 User Acceptance Testing (UAT)

UAT Conducted By	Panchami
Date	21-07-2026
Feedback Summary	Verified KPI totals, DAX measure outputs, and visual accuracy against the source dataset (diabetes.csv); confirmed navigation, filters, and drill-through behavior across all three report pages; no data discrepancies found
Sign-off Status	Approved (self-approved for portfolio release)

7.3 Performance Considerations

- Single flat table of 10,000 rows in Import mode — no performance issues expected at current scale
- Custom visuals (heat map matrix, radar charts) should be spot-checked for render time on lower-powered devices

8. DEPLOYMENT DOCUMENTATION

8.1 Current Status

The report currently exists only as a local Power BI Desktop file (Diabetes_Risk_Intelligence.pbix) and has not yet been published to a Power BI Service workspace.

8.2 Recommended Deployment Pipeline

Dev Workspace	My Workspace (Power BI Desktop, local .pbix) — active development and testing
Test/UAT Workspace	N/A — self-tested locally before publishing (see Section 7.2 UAT)
Production Workspace	Panchami – Portfolio Workspace

8.3 Publishing Steps

- Publish the .pbix to the target Power BI Service workspace
- Configure the data source credentials and (if needed) an On-premises Data Gateway for the CSV file location
- Set up a scheduled refresh policy matching source update frequency
- Assign workspace roles (Admin / Member / Contributor / Viewer)
- Share the report link or embed it in the relevant portal for stakeholders

8.4 Version History / Change Log

Version	Date	Changed By	Description
1.0	21-07-2026	Panchami	Completed all three report pages (Executive Overview, Risk Analysis, Demographics & Lifestyle); added custom visuals, navigation, and final formatting

9. MAINTENANCE & SUPPORT

9.1 Refresh Troubleshooting

- Confirm the diabetes.csv file path is still accessible if refresh fails
- If migrating to Power BI Service, verify gateway connectivity and credential validity

9.2 Ownership & Contacts

Report Owner	Panchami
Technical Contact	Panchaminaik930@gmail.com
Business Contact	Panchaminaik930@gmail.com/https://www.linkedin.com/in/panchami1578

9.3 Known Issues & Limitations

- The dataset is synthetic/simulated — not sourced from real clinical records — and should not be used for actual medical decisions without validation
- A synthetic DATE column was generated in Power Query for potential trend analysis but was removed in the final ETL step; the model currently has no time dimension, so period-over-period trending is not available
- Residence (Urban/Rural) is derived arbitrarily from Patient Id modulo 10 rather than genuine geographic data — treat this field as illustrative only
- No table relationships are defined; the model is a single flat table, which is adequate at this scale but should be revisited if additional fact/dimension tables are introduced
- Diabetes Risk Score is a simple rules-based weighted formula (40/30/20/10 weighting), not a statistically validated or ML-trained risk model

9.4 Planned Enhancements

- Reintroduce a proper date dimension if real longitudinal data becomes available
- Replace the rule-based Diabetes Risk Score with a validated statistical/ML model
- Add RLS if the report is rolled out to multiple regional or clinical teams

10. APPENDIX

10.1 Glossary of Terms

Term	Definition
BMI	Body Mass Index — $\text{weight(kg)/height(m)}^2$; Obese classified at $\text{BMI} \geq 30$ in this model
Glucose Level	Blood glucose concentration (mg/dL); High Glucose Rate flags patients above 180 mg/dL
IsHighRisk	Binary flag: 1 when a patient has both Glucose > 180 and BMI > 30
Risk Category	Low Risk (<30), Moderate Risk (30–59), High Risk (≥ 60) based on the Diabetes Risk Score
Lifestyle Group	Active / Moderate / Sedentary, derived from the physical_activity field

10.2 Full DAX Measure Listing (Verbatim)

The complete, unabridged DAX expression for every measure in the model is reproduced below for technical reference and audit purposes.

```
[_Measures].Total Patients
COUNTROWS(diabetes)
```

```
[_Measures].Diabetes Patients
CALCULATE(
    COUNTROWS(diabetes),
    diabetes[Diabetes Status] = "Diabetic"
)
```

```
[_Measures].Diabetes Prevalence
DIVIDE(
    [Diabetes Patients],
    [Total Patients],
    0
)
```

```
[_Measures].Average Glucose level
AVERAGE(Diabetes[Glucose Level])
```

```
[_Measures].Average BMI
AVERAGE(Diabetes[BMI])
```

```
[_Measures].High Risk Patients
CALCULATE(COUNTROWS(diabetes), diabetes[IsHighRisk] = 1)
```

```
[_Measures].High Glucose Rate
DIVIDE(
    CALCULATE(
        COUNTROWS(Diabetes),
```

```

Diabetes[Glucose Level]>180
),
[Total Patients]
)

```

```

[_Measures].Obesity Rate
DIVIDE(
  CALCULATE(
    COUNTROWS(Diabetes),
    Diabetes[BMI]>30
  ),
  [Total Patients]
)

```

```

[_Measures].Family History Rate
DIVIDE(
  CALCULATE(
    COUNTROWS(diabetes),
    diabetes[family_history] = 1
  ),
  [Total Patients],
  0
)

```

```

[_Measures].Smoking Rate
DIVIDE(
  CALCULATE(
    COUNTROWS(diabetes),
    diabetes[smoking] = 1
  ),
  COUNTROWS(diabetes),
  0
)

```

```

[_Measures].Active Lifestyle
DIVIDE(
  CALCULATE(COUNTROWS(diabetes), diabetes[Lifestyle Group] = "Active"),
  [Total Patients], 0
)

```

```

[_Measures].Sedentary Lifestyle
DIVIDE(
  CALCULATE(COUNTROWS(diabetes), diabetes[Lifestyle Group] = "Sedentary"),
  [Total Patients], 0)

```

```

[_Measures].Male Diabetes
DIVIDE(
  CALCULATE(
    COUNTROWS(Diabetes),
    Diabetes[Gender] = "Male",
    Diabetes[diabetes] = 1
  ),
  CALCULATE(
    COUNTROWS(Diabetes),
    Diabetes[diabetes] = 1
  )
)

```

```

    ),
    0
)

```

```

[_Measures].Female Diabetes
DIVIDE(
    CALCULATE(
        COUNTROWS(Diabetes),
        Diabetes[Gender] = "Female",
        Diabetes[diabetes] = 1
    ),
    CALCULATE(
        COUNTROWS(Diabetes),
        Diabetes[diabetes] = 1
    ),
    0
)

```

```

[_Measures].Diabetes Risk Score
ROUND(
    ([High Glucose Rate] * 100 * 0.40) +
    ([Obesity Rate] * 100 * 0.30) +
    ([Family History Rate] * 100 * 0.20) +
    ([Smoking Rate] * 100 * 0.10),
    0
)

```

```

[_Measures].Risk Category
SWITCH(
    TRUE(),
    [Diabetes Risk Score] < 30, "Low Risk",
    [Diabetes Risk Score] < 60, "Moderate Risk",
    "High Risk"
)

```

```

[_Measures].Risk Card Color
SWITCH(
    TRUE(),
    [Diabetes Risk Score] < 30, "#22C55E",    -- Green
    [Diabetes Risk Score] < 60, "#F59E0B",    -- Orange
    "#EF4444"                                -- Red
)

```

```

[Risk Factor].Risk Factor Impact
SWITCH(
    SELECTEDVALUE('Risk Factor'[Risk Factor]),
    "High Glucose", [High Glucose Rate],
    "Obesity", [Obesity Rate],
    "Family History", [Family History Rate],
    "Smoking", [Smoking Rate],
    BLANK()
)

```

```

[_Measures].Highest Lifestyle
VAR Activity =

```

```

TOPN(
    1,
    VALUES(diabetes[physical_activity]),
    [Diabetes Prevalence],
    DESC
)

RETURN
CONCATENATEX(Activity, diabetes[physical_activity])

```

```

[_Measures].Highest Lifestyle Insight
"The " &
[Highest Lifestyle] &
" activity group has the highest diabetes prevalence (" &
FORMAT(MAXX(
VALUES(diabetes[physical_activity]),
[Diabetes Prevalence]
),"0.00%") &
")."

```

```

[_Measures].Gender Insight
VAR Female = [Female Diabetes]
VAR Male = [Male Diabetes]

RETURN

IF(
Female>Male,

"Female diabetes prevalence is higher (" &
FORMAT(Female,"0.00%") &
") than male (" &
FORMAT(Male,"0.00%") &
").",

"Male diabetes prevalence is higher (" &
FORMAT(Male,"0.00%") &
") than female (" &
FORMAT(Female,"0.00%") &
")."
)

```

```

[_Measures].Family Insight
IF(

[Family History Rate]>0.5,

"More than 50% of diabetic patients have a family history.",

"Less than half of diabetic patients have a family history."
)

```

```

[_Measures].Residence Insight
VAR Urban =
    DIVIDE(
        CALCULATE(
            [Diabetes Patients],
            diabetes[Residence] = "Urban"

```

```
    ),
    [Diabetes Patients],
    0
  )

  VAR Rural =
    DIVIDE(
      CALCULATE(
        [Diabetes Patients],
        diabetes[Residence] = "Rural"
      ),
      [Diabetes Patients],
      0
    )

  RETURN
  IF(
    Urban > Rural,
    "Urban population contributes " &
    FORMAT(Urban, "0.00%") &
    " of diabetes cases.",
    "Rural population contributes " &
    FORMAT(Rural, "0.00%") &
    " of diabetes cases."
  )
)
```

10.3 Summary Statistics — Source Data (diabetes.csv, n = 10,000)

Field	Mean	Std Dev	Min	25%	Median	75%	Max
age	53.54	20.76	18	36	53	72	89
bmi	24.99	4.90	7.55	21.63	24.94	28.29	41.59
glucose	133.97	37.65	70	101	134	167	199
blood_pressure	129.48	29.04	80	104	129	155	179
cholesterol	209.34	52.14	120	164	210	255	299
heart_rate	89.70	17.14	60	75	90	104	119
sleep_hours	6.53	2.03	3	4.8	6.5	8.3	10
stress_level	5.48	2.88	1	3	5	8	10
diet_score	5.50	2.89	1	3	6	8	10
steps_per_day	10,502.64	5,503.03	1,000	5,688	10,527	15,271	19,999
work_hours	8.49	2.87	4	6	8	11	13
water_intake_ltr	2.99	1.16	1	2	3	4	5
insulin	156.01	83.77	10	84	157	229	299

10.4 Sample Data Records (first 5 rows)

Patient Id	age	gender	bmi	glucose	family_history	diabetes
1	69	Male	23.30	170	1	0
2	32	Male	25.00	184	0	0

Patient Id	age	gender	bmi	glucose	family_history	diabetes
3	89	Female	28.57	87	0	1
4	78	Male	15.85	96	1	1
5	38	Female	35.74	171	0	0

10.5 Reference Links

- Source data file: diabetes.csv
- Power BI file: Diabetes_Risk_Intelligence.pbix
- [Power BI workspace link, once published]

11. CONCLUSION

Project Achievements

- Consolidated 10,000 patient records spanning 20 clinical, demographic, and lifestyle attributes into a single, interactive Power BI model
- Designed and validated a transparent, rules-based Diabetes Risk Score (0–100) combining glucose, BMI, family history, and smoking behavior
- Delivered three purpose-built report pages — Executive Overview, Risk Analysis, and Demographics & Lifestyle — each targeted at a different level of analysis (headline KPIs, root-cause investigation, and population profiling)
- Built 26 DAX measures, including auto-generated narrative insight measures that translate the data into plain-language takeaways for non-technical stakeholders
- Documented the full data pipeline, model, and report design end-to-end to support handover, audit, and future enhancement

Business Value

The dashboard gives healthcare analysts and program administrators a fast, consistent way to identify where diabetes risk is concentrated and which factors are driving it, replacing ad hoc spreadsheet analysis with a governed, reusable model. The combined high-risk flag (glucose > 180 AND BMI > 30) narrows a broad population down to a specific, actionable 2.23% subgroup, allowing screening and outreach resources to be targeted rather than spread thinly across the entire patient base.

Power BI Techniques Used

- Power Query (M) transformations for data cleaning and derived-column engineering (AgeGroup, BMI Category, Risk Level, IsHighRisk, Lifestyle Group, Residence)
- DAX measures using CALCULATE, DIVIDE, SWITCH, TOPN, and CONCATENATEX for KPIs, rates, risk scoring, and dynamic narrative text
- Conditional formatting driven by a measure (Risk Card Color) to visually flag risk level on KPI cards
- Custom visuals (Tornado Chart, Lollipop Chart, Bullet Chart, Radar Chart, Heat Map Matrix, Radial Progress Ring) to extend beyond Power BI's native visual set
- Action-button page navigation for a guided, dashboard-style user experience across three report pages

Future Opportunities

- Replace the rules-based Diabetes Risk Score with a statistically validated or machine-learning-trained risk model
- Reintroduce a genuine date dimension to unlock trend, cohort, and period-over-period analysis
- Publish to a Power BI Service workspace with scheduled refresh, RLS, and a formal Dev/Test/Prod deployment pipeline
- Replace the synthetically derived Residence field with real geographic data for genuine urban/rural analysis
- Extend the model with real longitudinal patient data to validate current findings against a real clinical population

12. REFERENCES

The following resources were referenced or used in the development of this dashboard and its documentation.

- Microsoft Power BI Documentation — <https://learn.microsoft.com/power-bi/>
- Kaggle Dataset (dataset source) — <https://www.kaggle.com/> [exact dataset listing to be confirmed by project owner]
- Microsoft Learn — <https://learn.microsoft.com/>
- DAX Guide — <https://dax.guide/>
- Claude (Anthropic) — used to analyze the .pbix model and generate this documentation — <https://claude.ai>

End of Document